

# PARTIAL PROGRESS ON KRIVINE–STENGLE CERTIFICATES WITH UNIVERSAL QUANTIFIERS

**ABSTRACT.** Hu, Klep, and Nie asked for a non-Archimedean preordering theorem, a clean algebraic criterion for compactness or emptiness, and an analogue of the Krivine–Stengle Positivstellensatz for polynomial positivity sets defined by universal quantifiers. This note gives partial progress, not a complete solution. First, their perturbative Positivstellensatz implies that the compact-open closure of the integral quadratic module is exactly the cone of all polynomials nonnegative on the quantified feasible set. This yields exact closure criteria for emptiness and compactness, but not finitary algebraic certificates. Second, we construct a universally quantified set for which the literal Krivine–Stengle strict-positivity identity  $pf = 1 + q$  fails even when  $p$  and  $q$  are allowed to range over the saturated cone of all polynomials nonnegative on the set. Third, we define a natural tensor-integral preordering and exhibit an empty quantified feasible set for which  $-1$  does not belong to this preordering, although  $-1$  lies in its compact-open closure. Thus the classical identity and two natural preordering constructions do not extend in full generality, while other possible analogues remain open.

**Status.** The results below constitute *partial progress*. They give a complete negative answer to the narrow question of whether the classical strict-positivity identity  $pf = 1 + q$  extends verbatim to arbitrary closed quantifier sets, and they rule out natural preordering candidates. They do not provide, or rule out, every possible Krivine–Stengle-type theorem in the universally quantified setting.

## 1. INTRODUCTION

Let  $x = (x_1, \dots, x_n)$  and  $y = (y_1, \dots, y_m)$ , let  $Q \subseteq \mathbb{R}^m$  be closed, and let

$$g = (g_1, \dots, g_s) \in \mathbb{R}[x, y]^s.$$

The associated universally quantified feasible set is

$$(1) \quad K := \{x \in \mathbb{R}^n : g_j(x, y) \geq 0 \text{ for every } y \in Q \text{ and } j = 1, \dots, s\}.$$

Hu, Klep, and Nie [1] introduced a measure-based quadratic module for such sets and proved a Putinar-type theorem under an Archimedean hypothesis, together with a perturbative theorem without that hypothesis. They asked whether there is a preordering version of their theorem, whether compactness or emptiness of  $K$  has a clean algebraic reformulation, and whether a Krivine–Stengle Positivstellensatz exists in this setting.

For an ordinary basic closed semialgebraic set and its finitely generated preordering  $T$ , one strict-positivity form of the classical Krivine–Stengle theorem is

$$(2) \quad f > 0 \text{ on } K \iff \text{there exist } p, q \in T \text{ such that } pf = 1 + q;$$

see, for example, Stengle [3] or Marshall [2]. The present note investigates whether this identity, and natural preordering constructions modeled on the quantified data, can survive when  $Q$  is an arbitrary closed set.

Our conclusions are as follows.

- (i) The perturbative theorem of Hu–Klep–Nie immediately yields

$$\overline{\text{QM}[g, \nu]}^{\text{co}} = \text{Pos}(K),$$

where the closure is for uniform convergence on compact subsets of  $\mathbb{R}^n$  and  $\text{Pos}(K)$  is the saturated cone of polynomials nonnegative on  $K$ .

- (ii) There are data  $(Q, g, \nu)$  satisfying the hypotheses of Hu–Klep–Nie and a polynomial  $f > 0$  on  $K$  for which no  $p, q \in \text{Pos}(K)$  satisfy  $pf = 1 + q$ . Hence no sound polynomial preordering contained in  $\text{Pos}(K)$  can make the literal identity (2) complete in this generality.
- (iii) A natural tensor-integral preordering  $\mathcal{T}_{\otimes}[g, \nu]$ , which contains the quadratic module and allows products of constraints evaluated at independent quantifier variables, can fail to certify emptiness: for an explicit example,  $K = \emptyset$  but

$$-1 \notin \mathcal{T}_{\otimes}[g, \nu].$$

Nevertheless,  $-1$  is a compact-open limit of elements of the original quadratic module.

The first item is an exact topological characterization, but it is not a finitary algebraic certificate. The second and third items are obstructions to particular forms of a theorem, rather than a proof that every conceivable analogue must fail.

## 2. THE INTEGRAL QUADRATIC MODULE AND THE COMPACT-OPEN TOPOLOGY

Assume throughout this section that  $\nu$  is a finite nonnegative Borel measure on  $\mathbb{R}^m$  such that

$$\text{supp}(\nu) = Q,$$

all polynomial moments are finite, and the multivariate Carleman condition holds:

$$(3) \quad \sum_{d=1}^{\infty} \left( \int_{\mathbb{R}^m} |y_j|^{2d} d\nu(y) \right)^{-1/(2d)} = \infty, \quad j = 1, \dots, m.$$

If one of the moments in (3) is zero, the corresponding summand is interpreted as  $+\infty$ .

Set  $g_0 := 1$ . The integral quadratic module of Hu–Klep–Nie is

$$(4) \quad \text{QM}[g, \nu] := \left\{ \sum_{j=0}^s \int_Q \tau_j(x, y) g_j(x, y) d\nu(y) : \tau_j \in \Sigma^2[x, y] \right\} \subseteq \mathbb{R}[x].$$

Every element of  $\text{QM}[g, \nu]$  is nonnegative on  $K$ .

For  $R > 0$ , write

$$\|h\|_R := \sup_{\|x\|_2 \leq R} |h(x)|.$$

The compact-open topology on  $\mathbb{R}[x]$  is the locally convex topology generated by the seminorms  $\|\cdot\|_R$ ,  $R > 0$ . We write  $\overline{C}^{\text{co}}$  for the closure of a set  $C \subseteq \mathbb{R}[x]$  in this topology, and

$$\text{Pos}(K) := \{f \in \mathbb{R}[x] : f(x) \geq 0 \text{ for every } x \in K\}.$$

The cone  $\text{Pos}(K)$  is compact-open closed, since it is the intersection over  $x \in K$  of the inverse images of  $[0, \infty)$  under the continuous evaluation maps  $f \mapsto f(x)$ .

We use the following perturbative Positivstellensatz.

**Theorem 2.1** (Hu–Klep–Nie). *By [1, Corollary 3.7], for  $r \in \mathbb{N}$  define*

$$(5) \quad \Omega_r(x) := \sum_{j=1}^n \sum_{k=0}^r \frac{x_j^{2k}}{k!}.$$

*Then, for every  $f \in \mathbb{R}[x]$ ,*

$$f \in \text{Pos}(K) \iff \text{for every } \varepsilon > 0 \text{ there exists } r \in \mathbb{N} \text{ such that } f + \varepsilon \Omega_r \in \text{QM}[g, \nu].$$

## 3. EXACT COMPACT-OPEN CLOSURE

**Theorem 3.1.** *Under the assumptions above,*

$$(6) \quad \overline{\text{QM}[g, \nu]}^{\text{co}} = \text{Pos}(K).$$

*Proof.* Since every element of  $\text{QM}[g, \nu]$  is nonnegative on  $K$  and  $\text{Pos}(K)$  is compact-open closed,

$$\overline{\text{QM}[g, \nu]}^{\text{co}} \subseteq \text{Pos}(K).$$

For the reverse inclusion, let  $f \in \text{Pos}(K)$ . For each integer  $\ell \geq 1$ , apply Theorem 2.1 with  $\varepsilon = 1/\ell$  and choose  $r_\ell \in \mathbb{N}$  such that

$$f_\ell := f + \frac{1}{\ell} \Omega_{r_\ell} \in \text{QM}[g, \nu].$$

Fix  $R > 0$ . For  $\|x\|_2 \leq R$ ,

$$0 \leq \Omega_{r_\ell}(x) \leq \sum_{j=1}^n \sum_{k=0}^{\infty} \frac{R^{2k}}{k!} = ne^{R^2}.$$

Consequently,

$$\|f_\ell - f\|_R \leq \frac{ne^{R^2}}{\ell} \rightarrow 0.$$

Since  $R$  was arbitrary,  $f_\ell \rightarrow f$  in the compact-open topology. Thus  $f \in \overline{\text{QM}[g, \nu]}^{\text{co}}$ .  $\square$

Two consequences answer the compactness and emptiness questions at the level of compact-open closure.

**Corollary 3.2** (Emptiness in closure). *One has*

$$K = \emptyset \iff -1 \in \overline{\text{QM}[g, \nu]}^{\text{co}}.$$

*Proof.* If  $K = \emptyset$ , then  $\text{Pos}(K) = \mathbb{R}[x]$ , so the conclusion follows from Theorem 3.1. Conversely, if  $-1$  belongs to the closure, then Theorem 3.1 gives  $-1 \in \text{Pos}(K)$ , which is possible only when  $K$  is empty.  $\square$

**Corollary 3.3** (Compactness in closure). *The closed set  $K$  is compact if and only if there exists  $R \in \mathbb{R}$  such that*

$$R - \|x\|_2^2 \in \overline{\text{QM}[g, \nu]}^{\text{co}}.$$

*Proof.* If  $K$  is compact, then  $\|x\|_2^2$  is bounded above on  $K$ , so  $R - \|x\|_2^2 \in \text{Pos}(K)$  for some  $R$ . Apply Theorem 3.1. Conversely, membership in the closure implies, again by Theorem 3.1, that  $R - \|x\|_2^2 \geq 0$  on  $K$ . Hence  $K$  is bounded. It is closed by (1), and is therefore compact.  $\square$

**Remark 3.4.** *The criteria in Theorems 3.2 and 3.3 are exact but topological. Membership in a compact-open closure quantifies over an approximating sequence of certificates of unbounded degree. Thus these criteria do not by themselves supply the finitary algebraic reformulation sought in the open problem.*

## 4. FAILURE OF THE LITERAL KRIVINE–STENGLE IDENTITY

We first record that the measure hypothesis does not obstruct the use of arbitrary closed quantifier sets.

**Lemma 4.1** (Carleman measures on closed sets). *Every nonempty closed set  $Q \subseteq \mathbb{R}^m$  supports a finite nonnegative Borel measure  $\nu$  satisfying  $\text{supp}(\nu) = Q$ , having all polynomial moments finite, and satisfying the multivariate Carleman condition (3).*

*Proof.* Because  $Q$  is a separable metric space, choose a sequence  $(z_k)_{k \geq 1}$  dense in  $Q$ . Define

$$(7) \quad \nu := \sum_{k=1}^{\infty} 2^{-k} e^{-\|z_k\|_2^2} \delta_{z_k}.$$

This is a finite nonzero Borel measure. Its atoms are dense in  $Q$ , and all atoms lie in  $Q$ ; hence  $\text{supp}(\nu) = Q$ .

Fix a coordinate  $j$  and an integer  $d \geq 1$ . Since  $|z_{k,j}| \leq \|z_k\|_2$ ,

$$\begin{aligned} \int |y_j|^{2d} d\nu(y) &\leq \sum_{k=1}^{\infty} 2^{-k} e^{-\|z_k\|_2^2} \|z_k\|_2^{2d} \\ &\leq \sup_{t \geq 0} t^{2d} e^{-t^2} \sum_{k=1}^{\infty} 2^{-k} = \left(\frac{d}{e}\right)^d. \end{aligned}$$

The same bound shows that all polynomial moments are finite. Moreover,

$$\left( \int |y_j|^{2d} d\nu(y) \right)^{-1/(2d)} \geq \sqrt{\frac{e}{d}},$$

unless the moment is zero, in which case the left-hand side is  $+\infty$ . The series in (3) therefore diverges.  $\square$

The next elementary observation isolates the asymptotic obstruction.

**Lemma 4.2.** *Let  $K \subseteq \mathbb{R}^n$  and let  $f \in \mathbb{R}[x]$  be strictly positive on  $K$ . Suppose there is a sequence  $(x_k) \subseteq K$  such that*

$$(8) \quad p(x_k)f(x_k) \longrightarrow 0 \quad \text{for every } p \in \mathbb{R}[x].$$

*Then there do not exist  $p, q \in \text{Pos}(K)$  satisfying*

$$pf = 1 + q.$$

*Proof.* If such  $p, q$  existed, evaluating at  $x_k$  would give

$$p(x_k)f(x_k) = 1 + q(x_k) \geq 1$$

for every  $k$ , contradicting (8).  $\square$

We now realize the epigraph of an exponential function by a single polynomial inequality with a universal quantifier.

**Proposition 4.3.** *Let*

$$(9) \quad Q := \{(e^{-t}, e^{-t}(1+t)) : t \in \mathbb{R}\} \cup \{(0, 0)\} \subseteq \mathbb{R}^2,$$

*and define*

$$g(u, v, a, b) := v + au - b.$$

*Then  $Q$  is closed, and the universally quantified set*

$$K := \{(u, v) \in \mathbb{R}^2 : g(u, v, a, b) \geq 0 \text{ for every } (a, b) \in Q\}$$

*is exactly*

$$(10) \quad K = \{(u, v) \in \mathbb{R}^2 : v \geq e^{-u}\}.$$

*Moreover,  $Q$  supports a measure satisfying all hypotheses in Theorem 4.1.*

*Proof.* To prove that  $Q$  is closed, let

$$(a_k, b_k) = (e^{-t_k}, e^{-t_k}(1 + t_k))$$

converge in  $\mathbb{R}^2$ . If  $a_k \rightarrow a > 0$ , then  $t_k = -\log a_k \rightarrow -\log a$ , and the limit lies on the displayed curve. If  $a_k \rightarrow 0$ , then  $t_k \rightarrow +\infty$  and

$$b_k = e^{-t_k}(1 + t_k) \rightarrow 0,$$

so the limit is  $(0, 0)$ . Including sequences that already contain  $(0, 0)$  causes no change. Hence  $Q$  is closed.

For fixed  $u \in \mathbb{R}$ , the constraint corresponding to the curve point indexed by  $t$  is

$$v \geq h_u(t) := e^{-t}(1 + t - u).$$

A direct calculation gives

$$h'_u(t) = e^{-t}(u - t).$$

Thus  $h_u$  increases on  $(-\infty, u]$ , decreases on  $[u, \infty)$ , and has maximum

$$h_u(u) = e^{-u}.$$

The additional point  $(a, b) = (0, 0)$  imposes only  $v \geq 0$ , which is redundant because  $e^{-u} > 0$ . This proves (10). The final assertion follows from Theorem 4.1.  $\square$

**Theorem 4.4** (No saturated denominator identity). *For the data in Theorem 4.3, let*

$$f(u, v) := v.$$

*Then  $f > 0$  on  $K$ , but there do not exist  $p, q \in \text{Pos}(K)$  such that*

$$(11) \quad p(u, v)f(u, v) = 1 + q(u, v).$$

*Proof.* By (10),  $v \geq e^{-u} > 0$  on  $K$ , so  $f > 0$  on  $K$ . Consider the boundary points

$$x_u := (u, e^{-u}) \in K.$$

For an arbitrary polynomial  $p(u, v)$ , write

$$p(u, v) = \sum_{i=0}^d p_i(u)v^i, \quad p_i \in \mathbb{R}[u].$$

Then

$$(12) \quad p(u, e^{-u})e^{-u} = \sum_{i=0}^d p_i(u)e^{-(i+1)u} \rightarrow 0 \quad (u \rightarrow +\infty),$$

that is, the hypothesis of Theorem 4.2 holds along any sequence  $u \rightarrow +\infty$ . The conclusion follows.  $\square$

**Corollary 4.5.** *There is no rule assigning to every universally quantified data set  $(Q, g, \nu)$  a polynomial preordering  $T(Q, g, \nu) \subseteq \text{Pos}(K)$  such that, for every polynomial  $f$ ,*

$$f > 0 \text{ on } K \implies \exists p, q \in T(Q, g, \nu) \text{ with } pf = 1 + q.$$

*Proof.* Apply the asserted rule to the data of Theorem 4.3. Since  $T(Q, g, \nu) \subseteq \text{Pos}(K)$ , the resulting identity would contradict Theorem 4.4.  $\square$

**Remark 4.6.** *The obstruction in Theorem 4.4 concerns the literal polynomial identity  $pf = 1 + q$  with sound polynomial cones. It does not exclude certificates using localization, nonpolynomial multipliers, limits, auxiliary variables, extra hypotheses on  $Q$  or  $g$ , or a different algebraic formulation altogether.*

## 5. A NATURAL TENSOR-INTEGRAL PREORDERING

The quadratic module (4) does not include products of different quantified constraints. We next define a natural construction that does.

For  $r \geq 1$ , introduce independent quantifier variables

$$y^{(1)}, \dots, y^{(r)} \in \mathbb{R}^m.$$

For each  $a$ , let  $\alpha^{(a)} \in \{0, 1\}^s \setminus \{0\}$  and set

$$g^{\alpha^{(a)}}(x, y^{(a)}) := \prod_{j=1}^s g_j(x, y^{(a)})^{\alpha_j^{(a)}}.$$

**Definition 5.1** (Tensor-integral preordering). *The tensor-integral cone  $\mathcal{T}_{\otimes}[g, \nu] \subseteq \mathbb{R}[x]$  consists of all finite sums of terms of the following two types:*

- (a) *an SOS polynomial in  $x$ ;*
- (b) *for some  $r \geq 1$ , some  $\alpha^{(1)}, \dots, \alpha^{(r)} \in \{0, 1\}^s \setminus \{0\}$ , and some*

$$\sigma \in \Sigma^2[x, y^{(1)}, \dots, y^{(r)}],$$

*the polynomial*

$$(13) \quad \int_{Q^r} \sigma(x, y^{(1)}, \dots, y^{(r)}) \prod_{a=1}^r g^{\alpha^{(a)}}(x, y^{(a)}) d\nu^{\otimes r}.$$

All integrals are finite because the integrands are polynomials and  $\nu$  has all moments finite.

**Proposition 5.2.** *The cone  $\mathcal{T}_{\otimes}[g, \nu]$  is a preordering in  $\mathbb{R}[x]$ : it contains every square and is closed under addition and multiplication. It is sound,*

$$\mathcal{T}_{\otimes}[g, \nu] \subseteq \text{Pos}(K),$$

*and it contains the quadratic module,*

$$\text{QM}[g, \nu] \subseteq \mathcal{T}_{\otimes}[g, \nu].$$

*Consequently, the smallest polynomial preordering  $\text{PO}(\text{QM}[g, \nu])$  containing  $\text{QM}[g, \nu]$  also satisfies*

$$\text{PO}(\text{QM}[g, \nu]) \subseteq \mathcal{T}_{\otimes}[g, \nu].$$

*Proof.* Containment of squares and closure under addition are immediate from the definition. The product of two generator terms of the form (13) can, by Fubini's theorem, be written as a single integral over the product of their quantifier-variable spaces. The product of two SOS polynomials is SOS, since

$$\left( \sum_i a_i^2 \right) \left( \sum_j b_j^2 \right) = \sum_{i,j} (a_i b_j)^2.$$

Concatenating the independent quantifier variables and their exponent vectors therefore shows closure under multiplication. Products involving an SOS term are handled in the same way.

If  $x \in K$ , then every factor  $g_j(x, y^{(a)})$  in (13) is nonnegative for every  $y^{(a)} \in Q$ , while  $\sigma$  is nonnegative. Hence each generator is nonnegative at  $x$ , proving soundness.

For  $j \geq 1$ , a term

$$\int_Q \tau_j(x, y) g_j(x, y) d\nu(y)$$

from (4) is a tensor generator with  $r = 1$ . It remains to treat the  $j = 0$  term. We claim that

$$(14) \quad \int_Q \tau_0(x, y) d\nu(y) \in \Sigma^2[x] \quad \text{whenever } \tau_0 \in \Sigma^2[x, y].$$

Indeed, it suffices to consider  $\tau_0 = h(x, y)^2$ . Choose a finite monomial vector  $v(y)$  and a polynomial vector  $c(x)$  such that  $h(x, y) = c(x)^\top v(y)$ . The finite moment matrix

$$M := \int_Q v(y)v(y)^\top d\nu(y)$$

is positive semidefinite. Factoring  $M = B^\top B$  gives

$$\int_Q h(x, y)^2 d\nu(y) = c(x)^\top M c(x) = \|Bc(x)\|_2^2,$$

which is SOS in  $x$ . Summing proves (14), and hence  $\text{QM}[g, \nu] \subseteq \mathcal{T}_\otimes[g, \nu]$ . The final containment follows from the minimality of  $\text{PO}(\text{QM}[g, \nu])$ .  $\square$

Combining Theorems 3.1 and 5.2 gives an exact closure statement for this larger cone as well.

**Corollary 5.3.** *One has*

$$\overline{\mathcal{T}_\otimes[g, \nu]}^{\text{co}} = \text{Pos}(K).$$

*Proof.* By Theorem 5.2,

$$\text{QM}[g, \nu] \subseteq \mathcal{T}_\otimes[g, \nu] \subseteq \text{Pos}(K).$$

Take compact-open closures and apply Theorem 3.1 together with the compact-open closedness of  $\text{Pos}(K)$ .  $\square$

## 6. AN EMPTY SET NOT DETECTED BY THE TENSOR PREORDERING

We now show that passing from the quadratic module to the tensor-integral preordering does not, by itself, produce an exact algebraic emptiness certificate.

**Theorem 6.1.** *Let  $n = m = s = 1$ , let  $Q = [0, 1]$ , let  $\nu$  be Lebesgue measure on  $[0, 1]$ , and set*

$$(15) \quad g(x, y) := xy^2 - y = y(xy - 1).$$

*Then the universally quantified feasible set*

$$K = \{x \in \mathbb{R} : g(x, y) \geq 0 \text{ for every } y \in [0, 1]\}$$

*is empty, but*

$$(16) \quad -1 \notin \mathcal{T}_\otimes[g, \nu].$$

*In particular,*

$$-1 \notin \text{PO}(\text{QM}[g, \nu]).$$

*Nevertheless,*

$$-1 \in \overline{\text{QM}[g, \nu]}^{\text{co}} \subseteq \overline{\mathcal{T}_\otimes[g, \nu]}^{\text{co}}.$$

*Proof.* Fix  $x \in \mathbb{R}$ . Choose

$$0 < y < \min \left\{ 1, \frac{1}{|x| + 1} \right\}.$$

Then  $xy < 1$ , and therefore  $g(x, y) = y(xy - 1) < 0$ . Thus no  $x$  is feasible and  $K = \emptyset$ .

We next prove a leading-coefficient property. Consider a nonzero tensor generator with  $r \geq 1$ :

$$(17) \quad H(x) = \int_{[0,1]^r} \sigma(x, y_1, \dots, y_r) \prod_{a=1}^r (xy_a^2 - y_a) dy_1 \cdots dy_r,$$

where

$$\sigma = \sum_{\ell=1}^L h_{\ell}(x, y_1, \dots, y_r)^2$$

is nonzero. Let

$$d := \max_{1 \leq \ell \leq L} \deg_x h_{\ell}.$$

The coefficient of  $x^{2d}$  in  $\sigma$  is

$$(18) \quad C(y_1, \dots, y_r) := \sum_{\deg_x h_{\ell}=d} (\text{lc}_x h_{\ell}(y_1, \dots, y_r))^2.$$

This is a nonzero nonnegative polynomial. The coefficient of  $x^{2d+r}$  in the integrand in (17) is

$$C(y_1, \dots, y_r) \prod_{a=1}^r y_a^2.$$

It follows that the coefficient of  $x^{2d+r}$  in  $H$  is

$$(19) \quad \int_{[0,1]^r} C(y_1, \dots, y_r) \prod_{a=1}^r y_a^2 dy_1 \cdots dy_r > 0.$$

To justify strict positivity, observe that a nonzero polynomial cannot vanish on the open cube  $(0, 1)^r$ . Hence  $C$  is positive at some point of that cube and, by continuity, on a neighborhood of that point; the remaining factor in (19) is strictly positive there.

A nonzero SOS polynomial in the single variable  $x$  also has positive leading coefficient. Therefore every nonzero generator of  $\mathcal{T}_{\otimes}[g, \nu]$  has positive leading coefficient. In a finite sum of generators, the terms of maximal degree have positive leading coefficients, so they cannot cancel. Consequently every nonzero polynomial in  $\mathcal{T}_{\otimes}[g, \nu]$  has positive leading coefficient. The polynomial  $-1$  has negative leading coefficient, proving (16). Since  $\text{PO}(\text{QM}[g, \nu]) \subseteq \mathcal{T}_{\otimes}[g, \nu]$  by Theorem 5.2, the same nonmembership holds for the ordinary preordering generated by the quadratic module.

Finally, for  $N \geq 0$  define the SOS polynomial

$$(20) \quad \tau_N(y) := (2N+1)(2N+2)(1-y)^{2N}.$$

Then

$$\begin{aligned} H_N(x) &:= \int_0^1 \tau_N(y)(xy^2 - y) dy \\ &= (2N+1)(2N+2) \left( x \int_0^1 y^2(1-y)^{2N} dy - \int_0^1 y(1-y)^{2N} dy \right) \\ &= \frac{2}{2N+3}x - 1. \end{aligned}$$

Here we used the elementary beta-integral identities

$$\int_0^1 y(1-y)^{2N} dy = \frac{1}{(2N+1)(2N+2)}$$

and

$$\int_0^1 y^2(1-y)^{2N} dy = \frac{2}{(2N+1)(2N+2)(2N+3)}.$$

Each  $H_N$  belongs to  $\text{QM}[g, \nu]$ , and  $H_N \rightarrow -1$  uniformly on every compact subset of  $\mathbb{R}$ . This proves the closure assertion directly; it also follows from Theorem 3.2.  $\square$



**Remark 6.2.** *The tensor construction is deliberately generous: it allows arbitrary finite products of constraints evaluated at independent copies of the quantifier variable, together with joint SOS multipliers. Thus Theorem 6.1 rules out not only the original quadratic module, but also the ordinary preordering it generates and this larger natural tensor-integral preordering. It does not rule out every more elaborate construction.*

## 7. WHAT IS AND IS NOT RESOLVED

The preceding results separate three levels of the open question.

- (1) **Exact asymptotic characterization.** The equality

$$\overline{\text{QM}[g, \nu]}^{\text{co}} = \text{Pos}(K)$$

gives necessary and sufficient compact-open closure criteria for nonnegativity, emptiness, and compactness. This is a direct consequence of the existing perturbative theorem of Hu–Klep–Nie.

- (2) **Negative result for a verbatim extension.** The classical strict-positivity identity  $pf = 1 + q$  cannot hold for arbitrary closed quantifier sets, even with  $p$  and  $q$  drawn from the saturated cone  $\text{Pos}(K)$ . This completely answers that narrow formulation in the negative.
- (3) **Failure of natural preorderings.** Neither the ordinary preordering generated by  $\text{QM}[g, \nu]$  nor the larger tensor-integral preordering necessarily detects emptiness by membership of  $-1$ .

What remains open is the broader problem posed in [1]: there may exist a different finitary certificate, a theorem in a localized or enlarged algebra, or a complete result under additional structural hypotheses on  $Q$  and  $g$ . Accordingly, the present note should be read as partial progress and as a set of obstructions that any future analogue must overcome, not as a full solution.

## REFERENCES

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